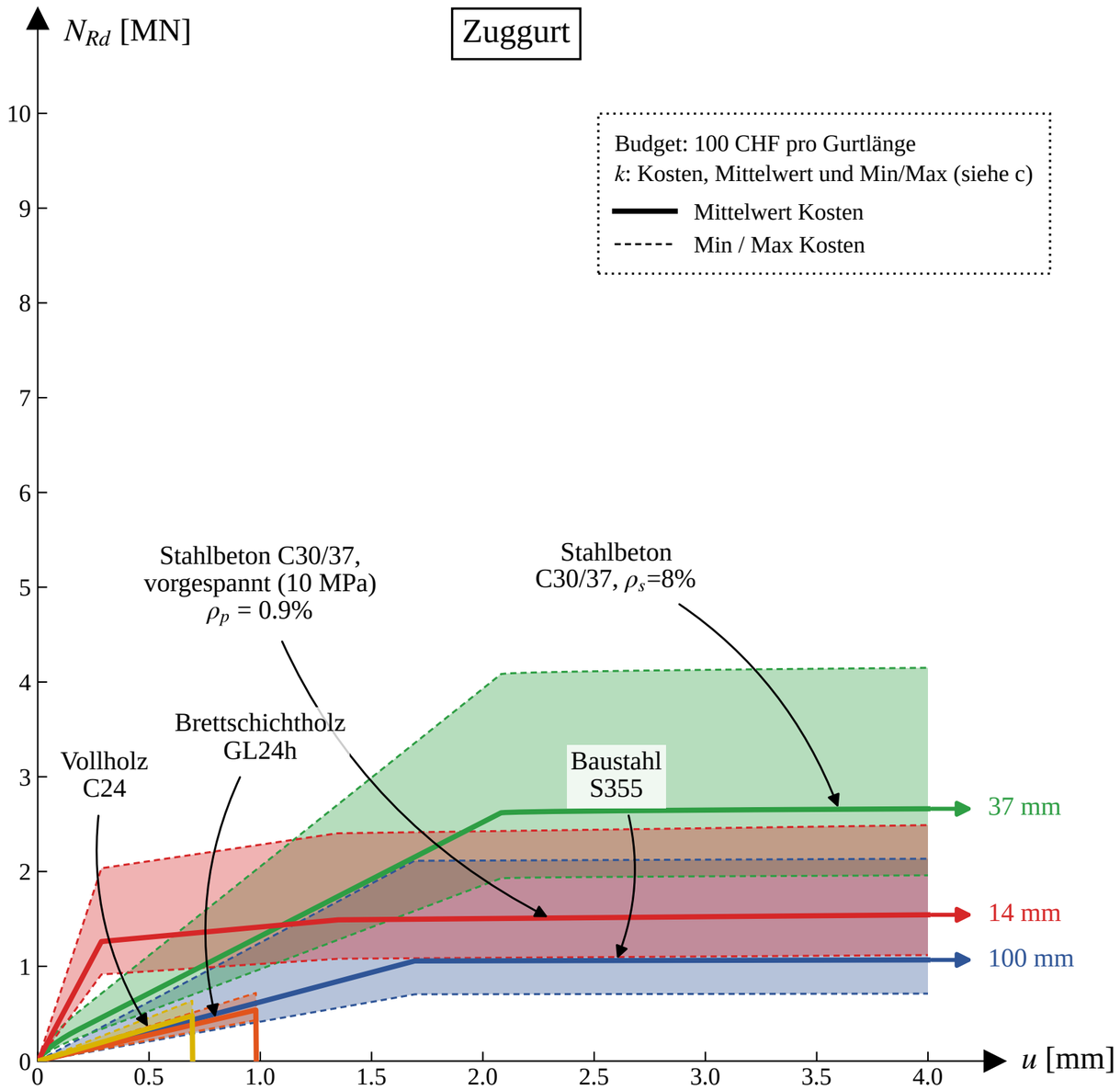
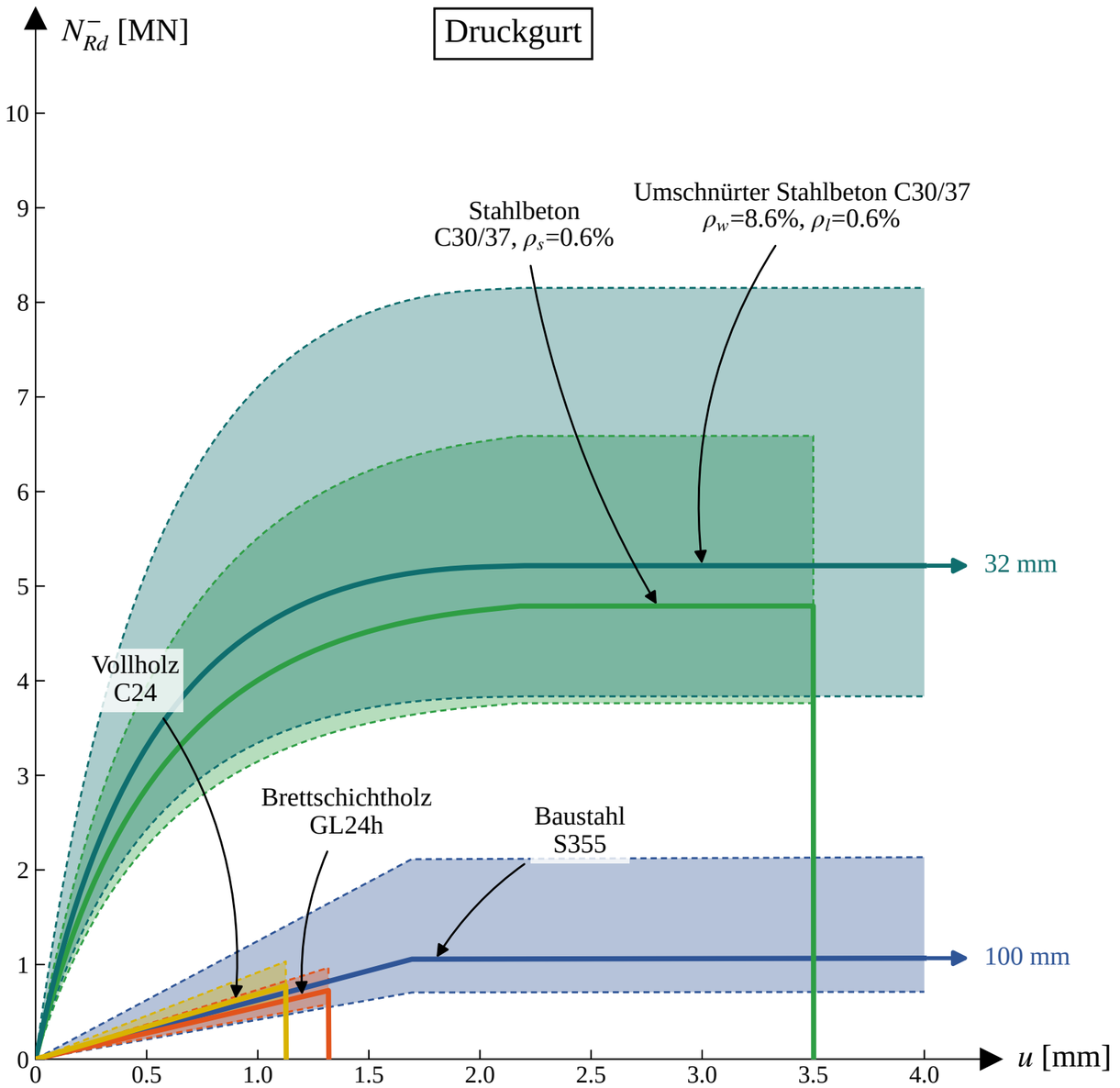
a)



b)



c)

vorgespannter Stahlbeton
σ_{cp} = 10 MPa
ρ_p = 0.9%
A_p = 1'084 mm^2
A_c = 119'944 mm^2
g_{0k} = 3.0 kN/m
k_c = 400 CHF/m^3, ±25% (160 CHF/t)
k_p = 48'000 CHF/m^3, ±50% (6'000 CHF/t)
Betonkosten inkl. Schalung

Baustahl
A = 3'125 mm^2
g_{0k} = 0.2 kN/m
k = 32'000 CHF/m^3, ±50% (4'000 CHF/t)
inkl. Korrosionsschutz

Stahlbeton C30/37
Zug: ρ_s = 8%
A = 75'301 mm^2
g_{0k} = 2.1 kN/m
Druck: ρ_s = 0.6%
A = 212'947 mm^2
g_{0k} = 5.2 kN/m
k_c = 400 CHF/m^3, ±25% (160 CHF/t)
k_s = 12'000 CHF/m^3, ±40% (1'500 CHF/t)
Betonkosten inkl. Schalung

Umschnürter Stahlbeton
C30/37, ρ_w = 8.6%, ρ_l = 0.6%
A = 66'357 mm^2
g_{0k} = 2.1 kN/m
k_c = 400 CHF/m^3, ±25% (160 CHF/t)
k_s = 12'000 CHF/m^3, ±40% (1'500 CHF/t)
Betonkosten inkl. Schalung

Brettschichtholz
GL24h
A = 50'000 mm^2
g_{0k} = 0.2 kN/m
k = 2'000 CHF/m^3, ±25% (4'000 CHF/t)
inkl. Montage + Verbinder

Vollholz
C24
A = 62'500 mm^2
g_{0k} = 0.3 kN/m
k = 1'600 CHF/m^3, ±25% (3'200 CHF/t)
inkl. Montage + Verbinder